Chapter 13: Experiment with Random Factors

When the **levels** of the **factors** used by the experimenter are the **specific levels** of **interest**, the experimenter is assuming that the factors in the experiment are **fixed factors**.

The implication of **fixed factors** is that statistical inferences and conclusions made about these factors are **confined** to the specific levels studied.

In some experimental situations, the **factor levels** are chosen at **random** from a **larger population** of **possible levels**, and the experimenter wishes to draw conclusions about the entire population of levels, not just those use in the experimental design. In this situation the factor is said to be a **random factor**.

We will discuss about random factors in this chapter.

Often **factor levels** are **chosen** at **random** from a larger population of potential levels, and we wish to make inferences about the entire population of levels.

Example 1: A drug company has its products manufactured in a large number of locations, and suspects that the purity of the product might vary from one location to another. **Three locations** are **randomly** chosen, and several samples of product from each are selected and tested for purity.

Statistical Model: For $i = 1, 2, \ldots, k$ and $j = 1,2, \ldots, n_i$, let

Response Variable

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Random Error

Overall Mean

Treatment Effect

Since the locations are randomly chosen, then their effects are random. Therefore, the τ_i are now random variables.

Statistical Model: For $i = 1, 2, \ldots, k$ and $j = 1,2, \ldots, n_i$, let

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Assumptions: Suppose τ_i and ε_{ij} are **independent** and

$$\tau_i \sim \mathrm{N}(0, \sigma_\tau^2), \quad \epsilon_{ij} \sim \mathrm{N}(0, \sigma_\varepsilon^2 = \sigma^2)$$

Note: $Var(y_{ij}) = \sigma_y^2 = \sigma_\tau^2 + \sigma^2$. The σ_τ^2 and σ^2 called **variance components** and the model is called a **random effects** model.

We would like to test

$$H_0: \sigma_\tau^2 = 0 \text{ versus } H_A: \sigma_\tau^2 > 0$$

Estimates:

$$\hat{\mu} = \bar{y}_{..} \text{ and } \hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$$

Statistical Model: For $i = 1,2,\ldots,k$ and $j = 1,2,\ldots,n$, let

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Define

$$SS_{Total} = \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{..})^2 ,$$

$$SS_{Treatments} = \sum_{i=1}^{k}\sum_{j=1}^{n}(\bar{y}_{i.} - \bar{y}_{..})^2$$

$$SS_{Error} = \sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i.})^2$$

It can be shown that

$$SS_{Total} = SS_{Treatments} + SS_{Error}$$

$$E(MS_{Treatments}) = \sigma^2 + n\boldsymbol{\sigma_\tau^2} \text{ and } E(MS_{Error}) = \sigma^2$$

Recall that for **fixed effects model** $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, we have

$$E\left(MS_{\text{Treatments}}\right) = \sigma^2 + \frac{n\sum_{i=1}^{k}\tau_i^2}{k-1}$$

For **random effects model** $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, we have

$$E(MS_{Treatments}) = \sigma^2 + n\boldsymbol{\sigma_\tau^2}$$

For both models,

$$E(MS_{Error}) = \sigma^2$$

The **ANOVA table**:

Source	df	SS	MS	F
Treatments	$k-1$	SSTr	$\mathrm{MSTr} = \frac{\mathrm{SSTr}}{\mathrm{k}-1}$	$\mathrm{F} = \frac{\mathrm{MSTr}}{\mathrm{MSE}}$
Error	$\mathrm{nk}-\mathrm{k}$	SSE	$\mathrm{MSE} = \frac{\mathrm{SSE}}{nk-k}$	
Total	$\mathrm{nk}-1$	SST		

Method of Moment Estimators: Since $E(MS_{Treatments}) = \sigma^2 + n\boldsymbol{\sigma_\tau^2}$ and $E(MS_{Error}) = \sigma^2$, we have

$$\sigma_\tau^2 = \frac{E(MS_{Treatments}) - E(MS_{Error})}{n}$$

Therefore, common (or **moments**) **estimator** of $\sigma^2$and σ_τ^2 are:

$$\hat{\sigma}^2 = MS_{Error}\,, \qquad \hat{\sigma}_\tau^2 = \frac{MS_{Treatments} - MS_{Error}}{n}$$

With unequal group sizes $n_1, n_2, \ldots, n_k$, we replace n in last equation by

$$n_0 = \frac{1}{k-1}\left[\sum_{i=1}^{k} n_i - \frac{\sum n_i^2}{\sum n_i}\right]$$

Note: Although the method of moment estimators are uniformly unbiased estimators, they have one unfortunate property. The estimate of σ_τ^2 **might** be **negative**.

Example 1: Apo A-I is known to help clear cholesterol from arteries. However, the apo genotype yields poor predictive values when screening for clinically defined atherosclerosis. This may be in part to difficulty in measurement.

The purpose of a study was to examine and quantify the variation among laboratories with respect to their measurement of an international references material for apo A-I and B (See Henderson, L. O., Hannon, W., Smith, S. J., and Cooper, G. R. (1987), *"An international collaborative study on standardization of apolipoproteins a –i and b "*, part ii, evaluation of contributions of antisers to among-laboratory variance components, Clinical Chemistry, **33**, 2250-2256)

Several measurements of the relative concentrations preparations of the reference material were made by 28 selected laboratories.

The table on the next page shows data from four representative labs.

Lab A	Lab B	Lab C	Lab D
1.195	1.155	1.021	1.163
1.144	1.173	1.037	1.171
1.167	1.171	1.022	1.182
1.249	1.175	1.064	1.184
1.177	1.153	1.094	1.175
1.217	1.139	0.992	1.134
1.187	1.185	1.072	1.169
	1.144		1.136

```
> Concen = c(1.195,1.144,1.167,1.249,1.177,1.217,1.187,
             1.155,1.173,1.171,1.175,1.153,1.139,1.185,1.144,
             1.021,1.037,1.022,1.064,1.094,0.992,1.072,
             1.163,1.171,1.182,1.184,1.175,1.134,1.169,1.136)

> Lab = as.factor(rep(1:4, c(7,8,7,8)))
> model = lm(Concen~Lab)
> anova(model)
```

R Codes to Retrieve the Mean Squares and Print Them:

```
> anova(model)

Analysis of Variance Table
Response: Concen
          Df    Sum Sq    Mean Sq    F value    Pr(>F)
Lab        3    0.092233  0.0307444  42.107     4.009e-10 ***
Residuals 26    0.018984  0.0007302

> model_anova = anova(model)

> ms = as.vector(model_anova[[3]])
> names(ms)=c("MS Lab", "MS Error")

> cat("Mean Square for Lab = ", ms[1], "\n",
      "Mean Square for Error = ", ms[2], "\n")
Mean Square for Lab =  0.03074443
Mean Square for Error =  0.0007301573
```

R Codes to Obtain Estimate of Variances and Print Them:

```
> n = c(7,8,7,8)
> k=4
> n0=(sum(n)-(sum(n^2)/sum(n)))/(k-1)
> sigma2t = (ms[1]-ms[2])/n0

> cat("Method of Moments Variance Component Estimates","\n",
   "Variance of Lab = ", sigma2t, "\n",
   "Variance of Error = ", ms[2], "\n")
```

Method of Moments Variance Component Estimates

Variance of Lab = 0.00400784

Variance of Error = 0.0007301573

Since $\hat{\sigma}^2 < \hat{\sigma}_\tau^2$, there is more variability among labs than within a lab.

Recall that $\hat{\sigma}_\tau^2 = 0.00400784$ and $\hat{\sigma}^2 = 0.0007301573$. Hence

$$\hat{\sigma}_y^2 = \hat{\sigma}_\tau^2 + \hat{\sigma}^2 = 0.00400784 + 0.0007301573 = 0.004737997$$

We estimate that 85% $\left(\frac{0.00400784}{0.004737997} = 0.85\right)$ of the variability in the concentration is attribute to variation between labs.

The **intera-class correlation coefficient** is defined by $ICC = \hat{\sigma}_\tau^2/(\hat{\sigma}_\tau^2 + \hat{\sigma}^2)$.

suppose $n_1 = n_2 = \cdots = n_k = n$. Let N be total number of observations and $L = \frac{1}{n}\left(\frac{MS(Treatments)}{MS(Error)}\frac{1}{F_{\frac{\alpha}{2},k-1,N-k}} - 1\right)$, U$= \frac{1}{n}\left(\frac{MS(Treatments)}{MS(Error)}\frac{1}{F_{1-\frac{\alpha}{2},k-1,N-k}} - 1\right)$.

It can be shown that a $100(1-\alpha)\%$ confidence interval for ICC is:

$$\left[\frac{L}{1+L}, \frac{U}{1+U}\right]$$

Maximum Likelihood and Restricted Maximum Likelihood Estimators: Although the method of moment estimators are uniformly unbiased estimators, they have one unfortunate property. The estimate of σ_τ^2 **might** be **negative**.

Maximum likelihood (**ML**) and **reduced** (or **restricted**) **maximum likelihood** (**REML**) are preferred methods of estimation that avoid this problem.

A desirable **property** of the **REML** estimates is that they are the **same** as the **method of moments** (analysis of variance estimates) when there is **equal replication** in each level of the random factor and $MS_{Treatments} > MS_{Eror}$.

The ML estimates and REML estimates are calculated by the R package **lme4** using a numerical solution for both unbalanced and balanced cases.

The **lmer** function in the R package **lme4** can compute the REML estimates of variance components.

Example 2: A manufacturer of packaged dry soup mixes was experiencing excessive variability in the package weights of a dry soup mix component called the “intermix”.

The intermix is a mixture of flavorful ingredients such as vegetable oil, salt, and so forth. Too much intermix in a soup packet gives it too strong a flavor, and not enough gives too week a flavor.

It was a two-step process to make the packaged soup mix. The first step was to make a large batch of soup and dry it on a rotary dryer. Next, the dried soup batch was placed into a mixer, where the intermix was added through ports as it was mixed. Then it was packaged in sealed bags of uniform weight.

There were several factors that could be changed in the first step (production of the dried soup batch), and several factors that could be changed in the second step (adding the intermix and mixing) that could possibly affect the variability of the weight of the intermix in each sealed bag.

Example 2 (Cont.): A factorial experiment was to be planned to find out which factors affected variability in intermix weights.

In order to determine which factors to include in a factorial experiment, a reasonable first step would be to partition the variability in intermix weight into the variability from soup batch to soup batch and the variability within a batch caused by the process to mix and add intermix to dried soup.

If there was little variability from batch to batch, the experiment would only need to consider factors involved in the mixing step.

In order to partition the variability in package weights into batch-to-batch and within batch, a sampling experiment was performed.

A random sample of **four** batches was selected over a month of production runs. From each batch, **three** 1-lb samples of the finished soup mix were taken from the mixing tank as it was being packaged. The weight of intermix was determine for each sample.

Example 2 (Cont.): The results of sampling experiment are shown in the following table.

Batch	Weight		
1	0.52	2.94	2.03
2	4.59	1.26	2.78
3	2.87	1.77	2.68
4	1.38	1.57	4.10

```
> Weight = c(0.52, 2.94, 2.03, 4.59, 1.26, 2.78,
             2.87, 1.77, 2.68,1.38, 1.57, 4.10)

> Batch = factor(rep(c(1:4), each = 3))
> model = lm(Weight~Batch)
> model_anova = anova(model)
```

Analysis of Variance Table
Response: Weight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Batch	**3**	**1.6606**	**0.55354**	**0.3197**	**0.8111**
Residuals	**8**	**13.8499**	**1.73123**		

R Codes to Obtain Estimate of Variances and Print Them:

```
> model_anova = anova(model)
> ms = as.vector(model_anova[[3]])
> n=3
> k=4
> sigma2t = (ms[1]-ms[2])/n

cat("Method of Moments Variance Component Estimates","\n",
    "Variance of Batch = ", sigma2t, "\n",
    "Variance of Error = ", ms[2], "\n")
```

Method of Moments Variance Component Estimates

Variance of Batch = -0.3925639

Variance of Error = 1.731233

Example 2 (Cont.): Since the estimate of σ_τ^2 is negative, we find its REML estimate.

```
> install.packages("lme4")

> library(lme4)
> model2 = lmer(Weight~1+(1|Batch))
```

The model formula **(Weight~1+(1|Batch))** for the function **lmer** is different than the model statement for **lm** or **aov** functions.

The response is on the left of the "**~**" operator as before, but the **1** on the right indicates that a **global mean** will be fit.

The vertical bar "**|**" in the second term of the model formula **separates** the fixed effects from the random effects.

The term to the **left** of the vertical bar is a **fixed effect** (the overall mean) and the term to the **right** (Batch) is a **random effect**.

R Codes to Obtain REML Estimate:

```
> library(lme4)
> model2 = lmer(Weight~1+(1|Batch))
> summary(model2)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: Weight ~ 1 + (1 | Batch)

REML criterion at convergence: 37.5

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.56147 -0.71722 -0.01614  0.43230  1.86604

Random effects:
 Groups   Name        Variance Std.Dev.
 Batch    (Intercept) 0.00     0.000
 Residual             1.41     1.187
Number of obs: 12, groups:  Batch, 4

Fixed effects:
            Estimate Std. Error t value
(Intercept)   2.3742     0.3428   6.926
```

An **asymptotic approximate confidence interval** for variance components can be obtained in the R package **lme4** using **likelihood profile method**.

The **profile** and **confint** functions in the R package **lme4** provide **confidence interval** on the **square root** (i.e. standard deviation) of the variance components.

Example 3: A dye manufacturer wanted to know if thee was an appreciable contribution to variability in dyestuff color yields owing to the quality of the intermediate acid batch used.

It was a two-step process. Intermediate acid batches were produced in one step, and in a later step the acid was used to produce the dyestuff.

The goal was to keep dyestuff color yield consistently high.

Example 3 (Cont.): If the majority of variation was caused by differences among the intermediate acid batches, then improvements efforts should be concentrated on the process that makes the acid batches.

If the majority of variation was within preparations of the dyestuff made from the same acid batch, improvement efforts should be focused on the process step of making the dyestuff.

A sampling experiment was run where six representative samples of H acid intermediate were taken from the step manufacturing process that produces.

From each acid sample, five preparations of the dyestuff Naphthalene 12B were made in a laboratory, and these were representative of the preparations that could be made with each sample.

The data from the sampling experiment is shown in the next page. The yields are given in grams of standard color.

Example 3 (Cont.): The yields are given in grams of standard color.

Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
1440	1490	1510	1440	1515	1445
1440	1495	1550	1445	1595	1450
1520	1540	1560	1465	1625	1455
1545	1555	1595	1545	1630	1480
1580	1560	1605	1595	1635	1520

```
> Yield =c(1440, 1440, 1520, 1545, 1580,
           1490, 1495, 1540, 1555, 1560,
           1510, 1550, 1560, 1595, 1605,
           1440, 1445, 1465, 1545, 1595,
           1515, 1595, 1625, 1630, 1635,
           1445, 1450, 1455, 1480, 1520)

> Sample = factor(rep(c(1:6), each = 5))
```

R Codes to Obtain ANOVA Table and Estimates of Variances:

```
> model = lm(Yield~Sample)
> anova(model)
```

Analysis of Variance Table

Response: Yield

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sample	**5**	**56358**	**11271.5**	**4.5983**	**0.004398 ****
Residuals	**24**	**58830**	**2451.3**		

```
> model_anova = anova(model)
> ms = as.vector(model_anova[[3]])
> names(ms)=c("MS Sample", "MS Error")
> cat("Mean Square for Lab = ", ms[1], "\n",
   "Mean Square for Error = ", ms[2], "\n")
```

Mean Square for Lab = 11271.5

Mean Square for Error = 2451.25

A 95% confidence interval for σ^2 is:

$$\left(\frac{SS_{Error}}{\chi^2_{\alpha/2,N-k}}, \frac{SS_{Error}}{\chi^2_{1-\alpha/2,N-k}}\right) = \left(\frac{SS_{Error}}{\chi^2_{0.025,24}}, \frac{SS_{Error}}{\chi^2_{0.975,24}}\right) = \left(\frac{58830}{39.364}, \frac{58830}{12.401}\right) = (1494.51, 4743.97)$$

A 95% confidence interval for σ is $\left(\sqrt{1494.51}, \sqrt{4743.97}\right) = (38.66, 68.88)$.

```
> df = as.vector(model_anova[[1]])
> ss = as.vector(model_anova[[2]])
> alpha = 0.05
> cat("\n","A ", 1-alpha, "%", "confidence interval for sigma is:","\n",
  "(", sqrt(ss[2]/qchisq(1-alpha/2,df[2])), " , ",
     sqrt(ss[2]/qchisq(alpha/2,df[2])) , ")", "\n")
```

A 0.95 % confidence interval for sigma is:

(38.65889 , 68.87608)

Let $F_O = \frac{MS(Treatments)}{MS_{Error}}$. Then an approximate 90% confidence interval σ_τ^2 is:

$$\left(\frac{SS_{Treatments}\left(1-\frac{F_{\alpha/4,k-1,N-k}}{F_O}\right)}{n\chi^2_{\alpha/4,k-1}}, \frac{SS_{Treatments}\left(1-\frac{F_{1-\alpha/4,k-1,N-k}}{F_O}\right)}{n\chi^2_{1-\alpha/4,k-1}} \right)$$

$$= \left(\frac{SS_{Treatments}\left(1-\frac{F_{0.025,5,24}}{F_O}\right)}{n\chi^2_{0.025,5}}, \frac{SS_{Treatments}\left(1-\frac{F_{0.975,5,24}}{F_O}\right)}{n\chi^2_{0.975,5}} \right)$$

$$= \left(\frac{56358\left(1-\frac{3.15482}{4.5983}\right)}{5(12.8325)}, \frac{56358\left(1-\frac{0.15929}{4.5983}\right)}{5(0.83121)} \right) = (275.732, 13090.723)$$

An approximate 90% confidence interval for σ_τ is

$$\left(\sqrt{275.732}, \sqrt{13090.723}\right) = (16.6, 114.4)$$

R Codes to Obtain an Approximate Confidence Interval for σ_τ^2:

```
> F= model_anova[[4]][1]

> alpha = 0.10
> n=5

> cat("\n","A ", 1-alpha, "%", "confidence interval for sigma of tau is:","\n",
  "(", sqrt(ss[1]*(1-qf(1-alpha/4,df[1],df[2])/F)/(n*qchisq(1- alpha/4,df[1]))),
    " , ",
   sqrt(ss[1]*(1-qf(alpha/4,df[1],df[2])/F)/(n*qchisq(alpha/4,df[1]))) , ")",
   "\n")
```

A 0.9 % confidence interval for sigma of tau is:
(16.605 , 114.4141)

Recall $F_O = \frac{MS(Treatments)}{MS_{Error}}$. Hence

$$L = \frac{1}{n}\left(\frac{MS(Treatments)}{MS(Error)}\frac{1}{F_{\frac{\alpha}{2},k-1,N-k}} - 1\right) = \frac{1}{n}\left(\frac{F_O}{F_{\frac{\alpha}{2},k-1,N-k}} - 1\right)$$

$$\text{U}= \frac{1}{n}\left(\frac{MS(Treatments)}{MS(Error)}\frac{1}{F_{1-\frac{\alpha}{2},k-1,N-k}} - 1\right) = \frac{1}{n}\left(\frac{F_O}{F_{1-\frac{\alpha}{2},k-1,N-k}} - 1\right)$$

Therefore, a 95% confidence interval for $\frac{\sigma_\tau^2}{\sigma^2}$ is:

$$\left(\frac{1}{n}\left(\frac{F_O}{F_{0.025,5,24}} - 1\right), \frac{1}{n}\left(\frac{F_O}{F_{0.975,5,24}} - 1\right)\right) = \left(\left(\frac{1}{5}\left(\frac{4.5983}{3.15482} - 1\right), \frac{1}{5}\left(\frac{4.5983}{0.15929} - 1\right)\right)\right) = (0.09,5.57)$$

A 95% confidence interval for $\frac{\sigma_\tau^2}{\sigma_\tau^2+\sigma^2}$ is:

$$\left(\frac{L}{1+L}, \frac{U}{1+U}\right) = \left(\frac{0.09}{1+0.09}, \frac{5.57}{1+5.57}\right) = (0.083,0.848)$$

R Codes to Obtain an Approximate Confidence Interval for $\frac{\sigma_\tau^2}{\sigma^2}$:

```
> alpha=0.05; n=5; k=6

> L = (1/n)*(F/qf(1-alpha/2,df[1],df[2])-1)
> U = (1/n)*(F/qf(alpha/2,df[1],df[2])-1)

> cat("\n","A ", 1-alpha, "%", "confidence interval for sigma2_tau/sigma2
      is:","\n", "(", L, " , ",U , ")", "\n")
```

A 0.95 % confidence interval for sigma2_tau/sigma2 is:
(0.09150769 , 5.57362)

R Codes to Obtain an Approximate Confidence Interval for $\frac{\sigma_\tau^2}{\sigma_\tau^2+\sigma^2}$:

```
> cat("\n","A ", 1-alpha, "%", "confidence interval ICC is:","\n",
        "(", L/(1+L), " , ",U/(1+U) , ")", "\n")
```

A 0.95 % confidence interval ICC is:
(0.08383605 , 0.8478768)

R Codes to Obtain an Approximate Confidence Interval for $\frac{\sigma_\tau^2}{\sigma_\tau^2+\sigma^2}$:

```
> install.packages("ICC")

> library(ICC)
> ICCest(Sample,Yield)
```

```
$ICC
[1] 0.4184874

$LowerCI
[1] 0.08383605

$UpperCI
[1] 0.8478768

$N
[1] 6

$k
[1] 5

$varw
[1] 2451.25

$vara
[1] 1764.05
```

Recall that the **moments estimator** of σ^2 and σ_τ^2 are:

$$\hat{\sigma}_\tau^2 = \frac{MS_{Treatments} - MS_{Error}}{n} = \frac{11271.5 - 2451.3}{5} = 1764.04$$

$$\hat{\sigma}^2 = MS_{Error} = 2451.3$$

R Codes to Obtain REML Estimate:

```
> library(lme4)
> model2 = lmer(Yield~1+(1|Sample))
> summary(model2)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: Yield ~ 1 + (1 | Sample)

REML criterion at convergence: 319.7

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.4117 -0.7634  0.1418  0.7792  1.8296

Random effects:
 Groups   Name        Variance Std.Dev.
 Sample   (Intercept) 1764     42.00
 Residual             2451     49.51
Number of obs: 30, groups:  Sample, 6

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1527.50      19.38    78.8
```

R Codes to Obtain ML Estimate:

```
> library(lme4)
> model2 = lmer(Yield~1+(1|Sample), REML = FALSE)
> summary(model2)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: Yield ~ 1 + (1 | Sample)

     AIC      BIC   logLik deviance df.resid
   333.3    337.5   -163.7    327.3       27

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.4315 -0.7972  0.1480  0.7721  1.8037

Random effects:
 Groups   Name        Variance Std.Dev.
 Sample   (Intercept) 1388     37.26
 Residual             2451     49.51
Number of obs: 30, groups:  Sample, 6

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1527.50      17.69   86.33
```

```
> anova(model)
```

Analysis of Variance Table

Response: Yield

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sample	**5**	**56358**	**11271.5**	**4.5983**	**0.004398 ****
Residuals	**24**	**58830**	**2451.3**		

Note that the **moments estimator** of σ and σ_τ are:

$$\hat{\sigma}_\tau = \sqrt{\frac{MS_{Treatments} - MS_{Error}}{n}} = \sqrt{\frac{11271.5 - 2451.34}{5}} = 42.00$$

$$\hat{\sigma} = \sqrt{MS_{Error}} = \sqrt{2451.34} = 49.51,$$

These estimates are equal to REML estimates.

R Codes to Obtain asymptotic confidence Interval for Variance Components Using REML Estimates:

```
>  library(daewr)

> pr = profile(model2)
> confint(pr)
```

	2.5 %	97.5 %
.sig01	**12.19854**	**84.06305**
.sigma	**38.22998**	**67.65770**
(Intercept)	**1486.45150**	**1568.54849**

Using moment of estimators, an approximate 95% confidence interval for σ_τ is $(12.23, 134.10)$ and a 95% confidence interval for σ was $(38.66, 68.88)$.

The asymptotic confidence intervals will be reasonably close to the exact confidence intervals when the degrees of freedom for the term corresponding to the variance component being estimated is greater than 45.

Two-Factor Factorial with Random Factors: Suppose that we have two factors, A and B that both have a **large number** of **levels** that are of interest.

We choose at **random** a levels of factor A and b levels of factor Band arrange these factor levels in a factorial experimental design. Supposed the experiment is replicated n times.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

Response Variable

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Random Error

Overall Mean

Effect of A

Effect of B

Interaction Effect

Since the levels of A and B are randomly chosen, then their effects are random. Therefore, the τ_i, β_j and $(\tau\beta)_{ij}$ are now random variables.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Assumptions: Suppose $\tau_i, \beta_j, (\tau\beta)_{ij}$ and ε_{ij} are all **independent** and

$$\tau_i \sim \mathrm{N}(0, \sigma_\tau^2), \qquad \beta_j \sim \mathrm{N}\left(0, \sigma_\beta^2\right), \qquad (\tau\beta)_{ij} \sim \mathrm{N}\left(0, \sigma_{(\tau\beta)}^2\right),$$

$$\epsilon_{ijk} \sim \mathrm{N}(0, \sigma_\varepsilon^2 = \sigma^2)$$

Note: $Var\left(y_{ij}\right) = \sigma_y^2 = \sigma_\tau^2 + \sigma_\beta^2 + \sigma_{(\tau\beta)}^2 + \sigma^2$. The σ_τ^2, $\sigma_\beta^2, \sigma_{(\tau\beta)}^2$ and σ^2 are called **variance components** and the model is called a **random effects** model.

We would like to test

$$H_0: \sigma_\tau^2 = 0 \text{ vs. } H_A: \sigma_\tau^2 > 0$$
$$H_0: \sigma_\beta^2 = 0 \text{ vs. } H_A: \sigma_\beta^2 > 0$$
$$H_0: \sigma_{(\tau\beta)}^2 = 0 \text{ vs. } H_A: \sigma_{(\tau\beta)}^2 > 0$$

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

The numerical calculations in the analysis of variance remain unchanged; that is SS_A, SS_B, SS_{AB}, and SS_{Error} are all calculated in the fixed effects case. However, the expected mean squares are:

$$E(MS_A) = \sigma^2 + n\sigma^2_{(\tau\beta)} + bn\sigma^2_\tau$$
$$E(MS_B) = \sigma^2 + n\sigma^2_{(\tau\beta)} + an\sigma^2_\beta$$
$$E(MS_{AB}) = \sigma^2 + n\sigma^2_{(\tau\beta)} \quad \text{and} \quad E(MS_{Error}) = \sigma^2$$

For testing $H_0: \sigma^2_{(\tau\beta)} = 0$ vs. $H_A: \sigma^2_{(\tau\beta)} > 0$, the test statistic is $F_O = \frac{MS_{AB}}{MS_{Error}}$.

For testing $H_0: \sigma^2_\tau = 0$ vs. $H_A: \sigma^2_\tau > 0$, the test statistic is $F_O = \frac{MS_A}{MS_{AB}}$.

For testing $H_0: \sigma^2_\beta = 0$ vs. $H_A: \sigma^2_\beta > 0$, the test statistic is $F_O = \frac{MS_B}{MS_{AB}}$.

Therefore, the usual ANOVA F-table for fixed model **cannot** be used.

Suppose all interactions are zero, then this imples the reduced model:

$$y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon_{ijk}.$$

In addition, we have

$$E(MS_A) = \sigma^2 + bn\sigma_\tau^2, \qquad E(MS_B) = \sigma^2 + an\sigma_\beta^2 \text{ and } \quad E(MS_{Error}) = \sigma^2$$

Therefore,

For testing $H_0: \sigma_{(\tau\beta)}^2 = 0$ vs. $H_A: \sigma_{(\tau\beta)}^2 > 0$, the test statistic is $F_O = \frac{MS_{AB}}{MS_{Error}}$.

For testing $H_0: \sigma_\tau^2 = 0$ vs. $H_A: \sigma_\tau^2 > 0$, the test statistic is $F_O = \frac{MS_A}{MS_{Error}}$.

For testing $H_0: \sigma_\beta^2 = 0$ vs. $H_A: \sigma_\beta^2 > 0$, the test statistic is $F_O = \frac{MS_B}{MS_{Error}}$.

So, the usual ANOVA F-test can be used.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

The expected mean squares are:

$$E(MS_A) = \sigma^2 + n\sigma^2_{(\tau\beta)} + bn\sigma^2_\tau$$
$$E(MS_B) = \sigma^2 + n\sigma^2_{(\tau\beta)} + an\sigma^2_\beta$$
$$E(MS_{AB}) = \sigma^2 + n\sigma^2_{(\tau\beta)} \quad \text{and} \quad E(MS_{Error}) = \sigma^2$$

The **moment estimators** of the variance components are:

$$\hat{\sigma}^2 = MS_{Error}, \qquad \hat{\sigma}^2_{(\tau\beta)} = \frac{MS_{AB} - MS_{Error}}{n}$$

$$\hat{\sigma}^2_\tau = \frac{MS_A - MS_{AB}}{bn}, \hat{\sigma}^2_\beta = \frac{MS_B - MS_{AB}}{an}$$

The **lmer** function in the R package **lme4** can compute the ML estimates and REML estimates of variance components.

Example 4: The purpose of a study was to the variability in measured features of manufactured products components. Assuming the gage or measuring instrument is properly calibrated, a measured value determined during a quality control inspection can be considered to be a function of the true feature dimension, the **gage repeatability**, and the **gage reproducibility**.

Gage repeatability is the ability of a single operator to obtain the **same measurement** value multiple times using the **same measuring instrument** (or gage) on the **same feature** of a single manufactured component (or part).

Gage reproducibility is the ability of **different operators** to obtain the **same measured value** multiple times using the **same gage** on the same parts.

If the variability in measurements caused by the gage repeatability plus the gage reproducibility is more than 10% of the tolerance range, the measurements may not be accurate enough to be used in monitoring product quality.

Example 4 (Cont.): A random sample of **ten manufactured parts** or components that are representative of the part-to-part variability in normal manufacturing was selected.

Next a random sample or a representative sample of **three** operator was selected.

Finally each inspector measures each part twice in a random order and the results are assembled in the following table.

Part	Operator 1		Operator 2		Operator 3		Part	Operator 1		Operator 2		Operator 3	
1	0.71	0.69	0.56	0.57	0.52	0.54	6	0.71	0.59	1.00	1.04	1.04	1.00
2	0.98	1.00	1.03	0.96	1.04	1.01	7	0.96	0.96	0.94	0.91	0.97	0.95
3	0.77	0.77	0.76	0.76	0.81	0.81	8	0.86	0.86	0.72	0.74	0.78	0.78
4	0.86	0.94	0.82	0.78	0.82	0.82	9	0.96	0.96	0.97	0.94	0.84	0.81
5	0.51	0.51	0.42	0.42	0.46	0.49	10	0.64	0.72	0.56	0.52	1.01	1.01

Statistical Model: For $i = 1, 2, \ldots, 10, j = 1,2,3$ and $k = 1,2$, let y_{ijk} denotes the kth measurement made by jth operator on the ith . Then

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

where $\boldsymbol{\tau_i}$ is the **part effect**, $\boldsymbol{\beta_j}$ is the **operator effect** and $(\boldsymbol{\tau\beta})_{ij}$ is the interaction effect.

Recall that $Var(y_{ij}) = \sigma_y^2 = \sigma_\tau^2 + \sigma_\beta^2 + \sigma_{(\tau\beta)}^2 + \sigma^2$. Therefore,

σ_τ^2 is the proportion of the total variance due to actual differences in par features.

σ_β^2 is the proportion of the total variance caused by differences among operators.

$\sigma_{(\tau\beta)}^2$ is the proportion of total variance caused by the interaction of operator and part.

Statistical Model: $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$

Recall that: $Var(y_{ij}) = \sigma_y^2 = \sigma_\tau^2 + \sigma_\beta^2 + \sigma_{(\tau\beta)}^2 + \sigma^2$

$\boldsymbol{\sigma^2}$ is the proportion of total variance caused by replicate measurements or **gage repeatability**.

The sum $\boldsymbol{\sigma_\beta^2 + \sigma_{(\tau\beta)}^2}$ is the **gage reproducibility**.

The reparability plus reproducibility $\boldsymbol{\sigma_\beta^2 + \sigma_{(\tau\beta)}^2 + \sigma^2}$ is a measure of the variance attribute to measurement error.

With **equal** number of **replicates** per subclass, we can use either **method of moments** or **REML** estimators of the variance components.

R Codes to Enter Data and Obtain ANOVA Table:

```
> y = c(0.71,0.69,0.98,1.00,0.77,0.77,0.86,0.94,0.51,0.51,
        0.71,0.59,0.96,0.96,0.86,0.86,0.96,0.96,0.64,0.72,
        0.56,0.57,1.03,0.96,0.76,0.76,0.82,0.78,0.42,0.42,
        1.00,1.04,0.94,0.91,0.72,0.74,0.97,0.94,0.56,0.52,
        0.52,0.54,1.04,1.01,0.81,0.81,0.82,0.82,0.46,0.49,
        1.04,1.00,0.97,0.95,0.78,0.78,0.84,0.81,1.01,1.01)

> Part = factor(rep(c(1:10), each = 2, times=3))
> Operator = factor(rep(c(1:3), each = 20))

> model = lm(y ~ Part+Operator+Part*Operator)
> anova(model)
```

> anova(model)

```
Analysis of Variance Table

Response: y
               Df  Sum Sq  Mean Sq F value    Pr(>F)
Part            9 1.44891 0.160991 214.178 < 2.2e-16 ***
Operator        2 0.02970 0.014852  19.758 3.353e-06 ***
Part:Operator  18 0.48393 0.026885  35.767 1.867e-15 ***
Residuals      30 0.02255 0.000752
---
```

Therefore, for testing $H_0: \sigma^2_{(\tau\beta)} = 0$ vs. $H_A: \sigma^2_{(\tau\beta)} > 0$, the test statistic is:

$$F_O = \frac{MS_{AB}}{MS_{Error}} = 35.767 \rightarrow p - value = P(F^{18}_{30} > 35.767) = 1.867 \times 10^{-15}.$$

For testing $H_0: \sigma^2_{\tau} = 0$ vs. $H_A: \sigma^2_{\tau} > 0$, the test statistic is:

$$F_O = \frac{MS_A}{MS_{AB}} = \frac{0.160991}{0.026885} = 5.988 \rightarrow p - value = P(F^{9}_{18} > 5.988) = 0.00064.$$

For testing $H_0: \sigma^2_{\beta} = 0$ vs. $H_A: \sigma^2_{\beta} > 0$, the test statistic is:

$$F_O = \frac{MS_B}{MS_{AB}} = \frac{0.014852}{0.026885} = 0.552 \rightarrow p - value = P(F^{2}_{18} > 0.522) = 0.60204.$$

> anova(model)

```
Analysis of Variance Table

Response: y
               Df  Sum Sq  Mean Sq F value    Pr(>F)
Part            9 1.44891 0.160991 214.178 < 2.2e-16 ***
Operator        2 0.02970 0.014852  19.758 3.353e-06 ***
Part:Operator  18 0.48393 0.026885  35.767 1.867e-15 ***
Residuals      30 0.02255 0.000752
---
```

Then the **moment estimators** of the variance components are:

$$\hat{\sigma}^2 = MS_{Error} = 0.000752,$$

$$\hat{\sigma}^2_{(\tau\beta)} = \frac{MS_{AB} - MS_{Error}}{n} = \frac{0.026885 - 0.000752}{2} = 0.01306565$$

$$\hat{\sigma}^2_{\tau} = \frac{MS_{A} - MS_{AB}}{bn} = \frac{0.160991 - 0.026885}{3(2)} = 0.022351$$

$$\hat{\sigma}^2_{\beta} = \frac{MS_{B} - MS_{AB}}{an} = \frac{0.014852 - 0.026885}{10(2)} = -0.00060165$$

R Codes to the Estimates of Variances:

```
> model_anova = anova(model)

> ms = as.vector(model_anova[[3]])

> n=2; a=10; b=3

> cat("Method of Moments Variance Component Estimate","\n",
      "Variance of Error = ", ms[4],"\n",
      "Variance of Interaction = ", (ms[3]-ms[4])/n, "\n",
      "Variance of Part = ", (ms[1]-ms[3])/(b*n), "\n",
      "Variance of Operator = ", (ms[2]-ms[3])/(a*n), "\n")
```

Method of Moments Variance Component Estimate
Variance of Error = 0.0007516667
Variance of Interaction = 0.01306667
Variance of Part = 0.02235093
Variance of Operator = -0.0006016667

A 95% confidence interval for σ^2 is:

$$\left(\frac{SS_{Error}}{\chi^2_{\alpha/2,ab(n-1)}}, \frac{SS_{Error}}{\chi^2_{1-\alpha/2,ab(n-1)}}\right) = \left(\frac{SS_{Error}}{\chi^2_{0.025,30}}, \frac{SS_{Error}}{\chi^2_{0.975,30}}\right) = \left(\frac{0.02255}{46.979}, \frac{0.02255}{16.791}\right) = (0.00048, 0.0134)$$

A 95% confidence interval for σ is $\left(\sqrt{0.00048}, \sqrt{0.0134}\right) = (0.022, 0.037)$.

```
> df = as.vector(model_anova[[1]])
> ss = as.vector(model_anova[[2]])
> alpha = 0.05
> cat("\n","A ", 1-alpha, "%", "confidence interval for sigma is:","\n",
  "(", sqrt(ss[4]/qchisq(1-alpha/2,df[4])), " , ",
     sqrt(ss[4]/qchisq(alpha/2,df[4])) , ")", "\n")
```

A 0.95 % confidence interval for sigma is:

(0.02190888 , 0.03664696)

Recall that, except mean square of error, the **expected value** of all other **mean squares** are **linear combinations** of two or more variance components.

Although exact **confidence intervals** can be obtained on **individual variance components** in the **balanced** two-factor design , they are **not** applicable to all designs or unbalanced data. However, approximate confidence intervals that are more generally applicable can be found.

In fact, whenever a variance component can be expressed in the form $\boldsymbol{\delta = c_1 E(MS_1) - c_2 E(MS_2)}$, where MS_1 and Ms_2 are mean squares in the ANOVA table and c_1 and c_2 are **positive constant**, the approximate confidence interval is applicable.

A function **vci** in the package **daewr** computes confidence intervals. The inputs for this function are **conf1 = 1-α**, **c1=** $\boldsymbol{c_1}$, **ms1 =** $\boldsymbol{MS_1}$, **nu1=** $\boldsymbol{df(MS_1)}$, **c2=** $\boldsymbol{c_2}$, **ms2 =** $\boldsymbol{MS_2}$, **nu2=** $\boldsymbol{df(MS_2)}$.

The expected mean squares are:

$$E(MS_A) = \sigma^2 + n\sigma^2_{(\tau\beta)} + bn\sigma^2_\tau$$

$$E(MS_B) = \sigma^2 + n\sigma^2_{(\tau\beta)} + an\sigma^2_\beta$$

$$E(MS_{AB}) = \sigma^2 + n\sigma^2_{(\tau\beta)} \quad \text{and} \quad E(MS_{Error}) = \sigma^2$$

Therefore,

$$\sigma^2_\tau = \frac{E(MS_A) - E(MS_{AB})}{bn} = \frac{1}{bn}E(MS_A) - \frac{1}{bn}E(MS_{AB})$$

$$\sigma^2_\beta = \frac{E(MS_B) - E(MS_{AB})}{an} = \frac{1}{an}E(MS_B) - \frac{1}{an}E(MS_{AB})$$

$$\sigma^2_{(\tau\beta)} = \frac{E(MS_{AB}) - E(MS_{Error})}{n} = \frac{1}{n}E(MS_{AB}) - \frac{1}{n}E(MS_{Error})$$

R Codes to Find Approximate Confidence Intervals for Variance Components:

```
> library(daewr)
> alpha=0.05; n=2; a=10; b=3
> vci(confl=1-alpha, c1 =1/(b*n), ms1=ms[1], nu1=df[1],
                          c2 =1/(b*n), ms2=ms[3], nu2=df[3] )
```

delta= 0.02235093 Lower Limit= 0.009214616 Upper Limit= 0.06817509

```
> vci(confl=1-alpha, c1 =1/(a*n), ms1=ms[2], nu1=df[2],
              c2 =1/(a*n), ms2=ms[3], nu2=df[3] )
```

delta= -0.0006016667 Lower Limit= -0.001937894 Upper Limit= 0.01314227

```
> vci(confl=1-alpha, c1 =1/n, ms1=ms[3], nu1=df[3],
                          c2 =1/n, ms2=ms[4], nu2=df[4] )
```

delta= 0.01306667 Lower Limit= 0.008002708 Upper Limit= 0.02539186

R Codes to Obtain REML Estimate:

```
> library(lme4)
> model2 = lmer(y~1+(1|Part)+(1|Operator)+(1|Part:Operator))
> summary(model2)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + (1 | Part) + (1 | Operator) + (1 | Part:Operator)

REML criterion at convergence: -133.9

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.43502 -0.36558 -0.01169  0.38978  1.94190

Random effects:
 Groups        Name        Variance  Std.Dev.
 Part:Operator (Intercept) 0.0124650 0.11165
 Part          (Intercept) 0.0225515 0.15017
 Operator      (Intercept) 0.0000000 0.00000
 Residual                  0.0007517 0.02742
Number of obs: 60, groups:  Part:Operator, 30; Part, 10; Operator, 3

Fixed effects:
            Estimate Std. Error t value
(Intercept)   0.7982     0.0518   15.41
```

R Codes to Obtain Confidence Intervals Using REML Estimate:

```
> library(daewr)
> pr = profile(model2)
> confint(pr)
```

	2.5 %	97.5 %
.sig01	0.08331198	Inf
.sig02	0.07963668	0.25120321
.sig03	0.00000000	0.09290403
.sigma	0.02171009	0.03614155
(Intercept)	0.69181745	0.90451588

A 95% confidence interval for $\sigma_{(\tau\beta)}$ is (0.083, ∞)

A 95% confidence interval for σ_{τ} is (0.079,0.251)

A 95% confidence interval for σ_{β} is (0.000,0.093)

A 95% confidence interval for σ is (0.022,0.036)

Using REML estimators , we found

A 95% confidence interval for $\sigma_{(\tau\beta)}$ is (0.083, ∞)

A 95% confidence interval for σ_{τ} is (0.079,0.251)

A 95% confidence interval for σ_{β} is (0.000,0.093)

A 95% confidence interval for σ is (0.022,0.036)

The confidence intervals for $\sigma, \sigma_{\tau}^2, \sigma_{\beta}^2$ and $\sigma_{(\tau\beta)}^2$ were obtained on pages 47 and 50. Therefore,

A 95% confidence interval for $\sigma_{(\tau\beta)}$ is

$$(\sqrt{0.008002708}, \sqrt{0.02539186}) = (0.089, 0.159).$$

A 95% confidence interval for σ_{τ} is

$$\left(\sqrt{0.009214616}, \sqrt{0.06817509}\right) = (0.096, 0.26).$$

A 95% confidence interval for σ_{β} is

$$(\sqrt{-0.001937894}, \sqrt{0.01314227}) = (\mathbf{NA}, 0.11).$$

A 95% confidence interval for σ is $(0.021, 0.037)$.

Recall that the **moment estimates** of the variance components are:

$$\hat{\sigma}^2 = 0.000752, \hat{\sigma}^2_{(\tau\beta)} = 0.01306565$$
$$\hat{\sigma}^2_{\tau} = 0.022351 \text{ and } \hat{\sigma}^2_{\beta} = -0.00060165$$

The **REML** estimates are:

$$\hat{\sigma}^2 = 0.0007517, \hat{\sigma}^2_{(\tau\beta)} = 0.0124650$$
$$\hat{\sigma}^2_{\tau} = 0.0225514 \text{ and } \hat{\sigma}^2_{\beta} = 0.000$$

From REML estimates, it can be seen that

$$\frac{\hat{\sigma}^2_{\beta} + \hat{\sigma}^2_{(\tau\beta)}}{\hat{\sigma}^2_{\beta} + \hat{\sigma}^2_{(\tau\beta)} + \hat{\sigma}^2} = \frac{0.000 + 0.0124650}{0.000 + 0.0124650 + 0.0007517} = 0.943$$

Therefore 94.3% of the measurement error is due to reproducibility. Therefore, to reduce measurement error, efforts should be concentrated on better training of operators rather than investing in more precise gages.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

The expected mean squares are:

$$E(MS_A) = \sigma^2 + n\sigma^2_{(\tau\beta)} + bn\sigma^2_\tau$$

$$E(MS_B) = \sigma^2 + n\sigma^2_{(\tau\beta)} + an\sigma^2_\beta$$

$$E(MS_{AB}) = \sigma^2 + n\sigma^2_{(\tau\beta)} \quad \text{and} \quad E(MS_{Error}) = \sigma^2$$

When there are an **unequal** number of replicates in each subclass, the methods of moments or analysis of variance estimators of the variance components are more **difficult** to compute.

The variance component estimates obtained by solving the equations that equate the mean squares to their expected values are **not** unique.

On the other hand, the **ML** estimators and **REML** estimators are unique and may be preferred in this situation.

Example 5: Consider a study to estimate the source of variability in an inter-laboratory assay of calcium in blood serum solutions with unknown concentrations. The data are given in the following table.

Laboratory	Standard Solution 1	2	3	4
A	87	92	179	177
		84	83	173
		80	76	166
B	80	69	138	151
		70	46	138
				132
C	70	67	173	176
		60	63	166
		44	48	

It should be mentioned that the estimated variance component for the **interaction** by the method of **moments estimator** is **negative**.

R Codes to Enter Data and Obtain ANOVA Table:

```
> y = c(87, 92, 179, 177,
            84,  83, 173,
            80,  76, 166,
        80, 69, 138, 151,
             70,  46, 138,
                      132,
        70, 67, 173, 176,
            60,  63, 166,
            44,  48        )

Lab = factor(rep(c("A","B","C"), c(10,8,9)))

Solution = factor(c(1:4,2:4,2:4,1:4,2:4,4,1:4,2:4,2:3))
```

R Codes to Obtain REML Estimate:

```
> library(lme4)
> model = lmer(y~1+(1|Lab)+(1|Solution)+(1|Lab:Solution))
> summary(model)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + (1 | Lab) + (1 | Solution) + (1 | Lab:Solution)

REML criterion at convergence: 265.2

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.64052 -0.55533 -0.06121  0.41731  2.33268

Random effects:
 Groups       Name        Variance Std.Dev.
 Lab:Solution (Intercept)    0.00   0.000
 Solution     (Intercept) 1493.72  38.649
 Lab          (Intercept)   28.03   5.294
 Residual                 1049.86  32.402
Number of obs: 27, groups:  Lab:Solution, 12; Solution, 4; Lab, 3

Fixed effects:
            Estimate Std. Error t value
(Intercept)   103.18      20.69   4.987
```

R Codes to Obtain Confidence Intervals Using REML Estimate:

> library(daewr)
> pr = profile(model)
> confint(pr)

```
> pr = profile(model)
Warning messages:
1: In nextpar(mat, cc, i, delta, lowcut, upcut) :
  Last two rows have identical or NA .zeta values: using minstep
2: In nextpar(mat, cc, i, delta, lowcut, upcut) :
  Last two rows have identical or NA .zeta values: using minstep
3: In nextpar(mat, cc, i, delta, lowcut, upcut) :
  Last two rows have identical or NA .zeta values: using minstep
4: In FUN(1:4[[4L]], ...) : non-monotonic profile for .sig01
>
>
> confint(pr)
                2.5 %    97.5 %
.sig01        0.00000  23.36823
.sig02       15.51759  85.77974
.sig03        0.00000  37.99254
.sigma       24.67321  44.91106
(Intercept) 57.46557 148.16540
Warning message:
In confint.thpr(pr) :
  bad spline fit for .sig01: falling back to linear interpolation
```

R Codes to Obtain ANOVA Table:

```
> model2 = lm(y ~ Lab+Solution+Lab*Solution)
> anova(model2)
```

```
Analysis of Variance Table

Response: y
             Df Sum Sq Mean Sq F value   Pr(>F)
Lab           2   2765  1382.4  0.9956 0.392633
Solution      3  35308 11769.2  8.4763 0.001567 **
Lab:Solution  6   1349   224.9  0.1620 0.983172
Residuals    15  20827  1388.5
```

Note that we still have $E(MS_{Error}) = \sigma^2$. Therefore, the estimate of σ^2 is $\hat{\sigma}^2 = MS_{Error} = 1388.5$.

With unequal replication the ANOVA mean squares are no longer distributed as multiples of chi-square random variables, so the confidence formula , obtained before by using chi-square distribution, are **not** appropriate.

R Codes to Obtain Confidence Intervals Using REML Estimate:

```
> library(daewr)
> pr = profile(model)
> confint(pr)
```

	2.5 %	97.5 %
.sig01	0.00000	23.36823
.sig02	15.51759	85.77974
.sig03	0.00000	37.99254
.sigma	24.67321	44.91106
(Intercept)	57.46557	148.16540

A 95% confidence interval for $\sigma_{(\tau\beta)}$ is (0.000, 23.368)

A 95% confidence interval for σ_{β} is (15.518, 85.779)

A 95% confidence interval for σ_{τ} is (0.000, 37.993)

A 95% confidence interval for σ is (24.673, 44.911)

Since there are only 3 labs, any asymptotic confidence interval for σ_τ^2 is **not** appropriate.

With unequal replication the ANOVA mean squares are no longer distributed as multiples of chi-square random variables, so the confidence formula , obtained before by using chi-square distribution, are **not** appropriate.

However, approximate confidence intervals can still be obtained using function **vci**.

Recall that whenever a variance component can be expressed in the form $\boldsymbol{\delta = c_1 E(MS_1) - c_2 E(MS_2)}$, where MS_1 and Ms_2 are mean squares in the ANOVA table and c_1 and c_2 are **positive constant**, the approximate confidence interval is applicable.

In fact, approximate confidence intervals can be obtained if **mean squares** from ANOVA table for **averages** and $\boldsymbol{n^* = ab/\sum(1/n_{ij})}$ instead of n are used.

R Codes to Obtain ANOVA Table Using Averages:

```
> y1 =tapply(y, list(Lab,Solution), mean)
> y1
      1        2                3                4
A    87   85.33333   112.66667     172.0000
B   80    69.50000     92.00000     140.3333
C   70    57.00000  94.66667       171.0000

> y1=c(y1)
 [1]  87.000  80.000   70.000   85.333   69.500  57.000 112.667
 [8]  92.000  94.667 172.000 140.333 171.000

> Lab1 = factor(rep(c("A","B","C"), time=4))
> Solution1 = factor(rep(c(1:4), each=3))

> model3 = lm(y1 ~ Lab1+Solution1+Lab1*Solution1)
> anova(model3)
```

Recall that the **moment estimator** of variance components for equal number of replicates are:

$$\sigma_\tau^2 = \frac{E(MS_A)-E(MS_{AB})}{bn} = \frac{1}{bn}E(MS_A) - \frac{1}{bn}E(MS_{AB})$$

$$\sigma_\beta^2 = \frac{E(MS_B)-E(MS_{AB})}{an} = \frac{1}{an}E(MS_B) - \frac{1}{an}E(MS_{AB})$$

$$\sigma_{(\tau\beta)}^2 = \frac{E(MS_{AB})-E(MS_{Error})}{n} = \frac{1}{n}E(MS_{AB}) - \frac{1}{n}E(MS_{Error})$$

Now, we should use mean squares from ANOVA table for **averages** and

$$n^* = \frac{ab}{\sum\left(\frac{1}{n_{ij}}\right)} = \frac{3(4)}{\left[\frac{1}{1}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{1}+\frac{1}{2}+\frac{1}{2}+\frac{1}{3}+\frac{1}{1}+\frac{1}{3}+\frac{1}{3}+\frac{1}{2}\right]}$$

$$= 1.846$$

> anova(model3)

```
Analysis of Variance Table

Response: y1
                 Df  Sum Sq Mean Sq F value Pr(>F)
Lab1              2   825.5   412.8
Solution1         3 15035.3  5011.8
Lab1:Solution1    6   624.5   104.1
Residuals         0     0.0
Warning message:
In anova.lm(model2) :
  ANOVA F-tests on an essentially perfect fit are unreliable
```

The estimates are:

$\hat{\sigma}^2 = MS(Error) = 1388.5$ from Original ANOVA table on page 60

$$\hat{\sigma}^2_{(\tau\beta)} = \frac{MS_{AB} - MS_{Error}}{n^*} = \frac{104.1 - 1388.5}{1.846} = -695.8$$

$$\hat{\sigma}^2_{\tau} = \frac{MS_A - MS_{AB}}{bn^*} = \frac{412.8 - 104.1}{4(1.846)} = 41.8$$

$$\hat{\sigma}^2_{\beta} = \frac{MS_B - MS_{AB}}{an^*} = \frac{5011.8 - 104.1}{3(1.846)} = 886.2$$

R Codes to Obtain Approximate Estimates:

```
> model_anova2 = anova(model2)
> model_anova3 = anova(model3)
> ms = as.vector(model_anova3[[3]])
> ms[4] = model_anova2[[3]][4]
> a=3; b=4; n=c(1,3,3,3,1,2,2,3,1,3,3,2)
> nstar=a*b/sum(1/n)
> cat("Method of Moments Variance Component Estimate","\n",
   "Variance of Error = ", ms[4],"\n",
   "Variance of Interaction = ", (ms[3]-ms[4])/nstar, "\n",
   "Variance of Part = ", (ms[1]-ms[3])/(b*nstar), "\n",
   "Variance of Operator = ", (ms[2]-ms[3])/(a*nstar), "\n")
```

Method of Moments Variance Component Estimate
Variance of Error = 1388.478
Variance of Interaction = -695.7099
Variance of Part = 41.79861
Variance of Operator = 886.1073

R Codes to Find Approximate Confidence Intervals for Variance Components:

```
> library(daewr)
> df = as.vector(model_anova3[[1]])
> df[4] = sum(n)-a*b     # we should adjust df for Error

> vci(confl=1-alpha, c1 =1/(b*nstar), ms1=ms[1], nu1=df[1],
              c2 =1/(b*nstar), ms2=ms[3], nu2=df[3] )
```

delta= 41.79861 Lower Limit= -8.618328 Upper Limit= 1075.628

```
> vci(confl=1-alpha, c1 =1/(a*nstar), ms1=ms[2], nu1=df[2],
              c2 =1/(a*nstar), ms2=ms[3], nu2=df[3] )
```

delta= 886.1073 Lower Limit= 331.4614 Upper Limit= 7696.809

```
> vci(confl=1-alpha, c1 =1/nstar, ms1=ms[3], nu1=df[3],
              c2 =1/nstar, ms2=ms[4], nu2=df[4] )
```

delta= -695.7099 Lower Limit= -1497.26 Upper Limit= -359.406

Confidence intervals using **REML** estimate are:

A 95% confidence interval for $\sigma_{(\tau\beta)}$ is (0.000, 23.368)

A 95% confidence interval for σ_{β} is (15.518, 85.779)

A 95% confidence interval for σ_{τ} is (0.000, 37.993)

A 95% confidence interval for σ is (24.673, 44.911)

Approximate confidence intervals using **vci** function are:

A 95% confidence interval for $\sigma_{(\tau\beta)}$ is

$$(\sqrt{-1497.26}, \sqrt{-359.406}) = (\mathbf{NA}, \mathbf{NA}).$$

A 95% confidence interval for σ_{β} is

$$(\sqrt{331.4614}, \sqrt{7696.809}) = (18.206, 87.731).$$

A 95% confidence interval for σ_{τ} is

$$\left(\sqrt{-8.618328}, \sqrt{1075.628}\right) = (\mathbf{NA}, 32.797).$$

Mixed Model: Suppose that we have two factors, A and B. Factor **A** is **fixed** and factor **B** is **random**. Suppose factor A has a levels and we choose at **random** b levels of factor B and arrange these factor levels in a factorial experimental design. Supposed the experiment is replicated n times.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Response Variable

Overall Mean

Effect of A

Effect of B

Interaction Effect

Random Error

Here τ_i is **fixed effect**. $\boldsymbol{\beta_j}$ and $(\boldsymbol{\tau\beta})_{ij}$ are now **random effects**.

This model is called mixed model.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Assumptions: Suppose $\beta_j, (\tau\beta)_{ij}$ and ε_{ij} are all **independent** and

$$\sum_{i=1}^{a} \tau_i = 0,$$

$$\beta_j \sim \mathrm{N}\left(0, \sigma_\beta^2\right), \quad (\tau\beta)_{ij} \sim \mathrm{N}\left(0, \frac{\boldsymbol{a-1}}{\boldsymbol{a}} \sigma_{(\tau\beta)}^2\right), \quad \epsilon_{ijk} \sim \mathrm{N}(0, \sigma_\varepsilon^2 = \sigma^2)$$

$$\sum_{i=1}^{a} (\tau\beta)_{ij} = 0.$$

Note: The last restriction implies that certain interaction elements at different levels of the fixed factor are **not** independent. In fact, it can be shown that

$$Cov\left((\tau\beta)_{ij}, (\tau\beta)_{i'j}\right) = -\frac{1}{a}\sigma_{(\tau\beta)}^2 \text{ for } i \neq i'$$

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

Assumptions: Suppose $\beta_j, (\tau\beta)_{ij}$ and ε_{ij} are all **independent** and

$$\sum_{i=1}^{a} \tau_i = 0, \sum_{i=1}^{a} (\tau\beta)_{ij} = 0.$$

$$\beta_j \sim \mathrm{N}\left(0, \sigma_\beta^2\right), \quad (\tau\beta)_{ij} \sim \mathrm{N}\left(0, \frac{\boldsymbol{a-1}}{\boldsymbol{a}} \sigma_{(\tau\beta)}^2\right), \quad \epsilon_{ijk} \sim \mathrm{N}(0, \sigma_\varepsilon^2 = \sigma^2)$$

Note: $Var\left(y_{ij}\right) = \sigma_y^2 = \sigma_\beta^2 + \sigma_{(\tau\beta)}^2 + \sigma^2$. The $\sigma_\beta^2, \sigma_{(\tau\beta)}^2$ and σ^2 are called **variance components** and the model is called a **mixed** model. We would like to test

$$H_0: \tau_i = 0 \text{ vs. } H_A: \tau_i \neq 0$$
$$H_0: \sigma_\beta^2 = 0 \text{ vs. } H_A: \sigma_\beta^2 > 0$$
$$H_0: \sigma_{(\tau\beta)}^2 = 0 \text{ vs. } H_A: \sigma_{(\tau\beta)}^2 > 0$$

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

The numerical calculations in the analysis of variance remain unchanged; that is SS_A, SS_B, SS_{AB}, and SS_{Error} are all calculated in the fixed effects case. However, the expected mean squares are:

$$E(MS_A) = \sigma^2 + n\sigma^2_{(\tau\beta)} + \frac{bn\sum \tau_i^2}{a-1}$$

$$E(MS_B) = \sigma^2 + an\sigma^2_{\beta}$$

$$E(MS_{AB}) = \sigma^2 + n\sigma^2_{(\tau\beta)} \quad \text{and} \quad E(MS_{Error}) = \sigma^2$$

For testing $H_0: \sigma^2_{(\tau\beta)} = 0$ vs. $H_A: \sigma^2_{(\tau\beta)} > 0$, the test statistic is $F_O = \frac{MS_{AB}}{MS_{Error}}$.

For testing $H_0: \tau_i = 0$ vs. $H_A: \tau_i \neq 0$, the test statistic is $F_O = \frac{MS_A}{\boldsymbol{MS_{AB}}}$.

For testing $H_0: \sigma^2_{\beta} = 0$ vs. $H_A: \sigma^2_{\beta} > 0$, the test statistic is $F_O = \frac{MS_B}{MS_{Error}}$.

Therefore, the usual ANOVA F-table for fixed model **cannot** be used, except when interactions are zero.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

The estimate of fixed effects are:

$$\hat{\mu} = \bar{y}_{\ldots}, \hat{\tau}_i = \bar{y}_{i..} - \bar{y}_{\ldots} \text{ for } i = 1,2, \ldots, a$$

The **moment estimators** of the variance components are:

$$\hat{\sigma}^2 = MS_{Error}, \qquad \hat{\sigma}^2_{(\tau\beta)} = \frac{MS_{AB} - MS_{Error}}{n} \quad \hat{\sigma}^2_{\beta} = \frac{MS_B - MS_{Error}}{an}$$

The **lmer** function in the R package **lme4** can compute the ML estimates and REML estimates of variance components.

Statistical Model: For $i = 1, 2, \ldots, a, j = 1,2, \ldots, b$ and $\mathrm{k} = 1,2, \ldots, n$, let

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$$

It can be shown that

$$E(\hat{\tau}_i - \hat{\tau}_{i'}) = E\left[(\bar{y}_{i..} - \bar{y}_{...}) - (\bar{y}_{i'..} - \bar{y}_{...})\right] = E(\bar{y}_{i..} - \bar{y}_{i'..}) = \tau_i - \tau_{i'}$$

$$Var(\hat{\tau}_i - \hat{\tau}_{i'}) = Var(\bar{y}_{i..} - \bar{y}_{i'..}) = \frac{2}{nb} MS_{AB}$$

Note that this is the standard error that we would use if this was a fixed effect model.

A $100(1-\alpha)\%$ confidence interval for $\tau_i - \tau_{i'}$ is:

$$\bar{y}_{i..} - \bar{y}_{i'..} \pm t^*_{\frac{\alpha}{2}, df(MS_{AB})} \sqrt{\frac{2}{nb} MS_{AB}}$$

For multiple comparisons using Tukey's method replace $t^*_{\frac{\alpha}{2}, df(MS_{AB})}$ by $qtukey(1-\alpha, a, df(MS_{AB}) = (a-1)(b-1))/\sqrt{2}$.

Example 6: It was assumed, in Example 4, that the 3 operators were chosen from a large pool of possible operators, about which we want to make inferences.

Suppose instead that the manufacturer employs only 3 operators, and wishes to make inferences only about these three.

Therefore, the **effect of operator** is now **fixed** and the **effect of part** is **random**.

The F-statistic for part and interaction in the ANOVA table are correct.

The F-statistic, and therefore the corresponding p-value, for operator in the ANOVA table is wrong and can be fixed by simple calculations.

R Codes to Enter Data and Obtain ANOVA Table:

```
> y = c(0.71,0.69,0.98,1.00,0.77,0.77,0.86,0.94,0.51,0.51,
        0.71,0.59,0.96,0.96,0.86,0.86,0.96,0.96,0.64,0.72,
        0.56,0.57,1.03,0.96,0.76,0.76,0.82,0.78,0.42,0.42,
        1.00,1.04,0.94,0.91,0.72,0.74,0.97,0.94,0.56,0.52,
        0.52,0.54,1.04,1.01,0.81,0.81,0.82,0.82,0.46,0.49,
        1.04,1.00,0.97,0.95,0.78,0.78,0.84,0.81,1.01,1.01)

> Part = factor(rep(c(1:10), each = 2, times=3))
> Operator = factor(rep(c(1:3), each = 20))

> model = lm(y ~ Operator+Part + Operator*Part)
> anova(model)
```

> anova(model)

```
Analysis of Variance Table

Response: y
               Df  Sum Sq  Mean Sq F value    Pr(>F)
Operator        2 0.02970 0.014852  19.758 3.353e-06 ***
Part            9 1.44891 0.160991 214.178 < 2.2e-16 ***
Operator:Part  18 0.48393 0.026885  35.767 1.867e-15 ***
Residuals      30 0.02255 0.000752
```

Therefore, for testing $H_0: \sigma^2_{(\tau\beta)} = 0$ vs. $H_A: \sigma^2_{(\tau\beta)} > 0$, the test statistic is:

$$F_O = \frac{MS_{AB}}{MS_{Error}} = 35.767 \rightarrow p - value = P(F^{18}_{30} > 35.767) = 1.867 \times 10^{-15}.$$

For testing $H_0: \sigma^2_{\beta} = 0$ vs. $H_A: \sigma^2_{\beta} > 0$, the test statistic is:

$$F_O = \frac{MS_B}{MS_{Error}} = 214.178 \rightarrow p - value = P(F^{9}_{30} > 214.178) = 2.2 \times 10^{-16}.$$

For testing $H_0: \tau_i = 0$ vs. $H_A: \tau_i \neq 0$, the test statistic is:

$$F_O = \frac{MS_A}{MS_{AB}} = \frac{0.014852}{0.026885} = 0.552 \rightarrow p - value = P(F^{2}_{18} > 0.522)$$

$$= \mathbf{0.60204}.$$

> anova(model)

```
Analysis of Variance Table

Response: y
               Df  Sum Sq  Mean Sq F value    Pr(>F)
Operator        2 0.02970 0.014852  19.758 3.353e-06 ***
Part            9 1.44891 0.160991 214.178 < 2.2e-16 ***
Operator:Part  18 0.48393 0.026885  35.767 1.867e-15 ***
Residuals      30 0.02255 0.000752
```

Then the **moment estimators** of the variance components are:

$$\hat{\sigma}^2 = MS_{Error} = 0.000752,$$

$$\hat{\sigma}^2_{(\tau\beta)} = \frac{MS_{AB} - MS_{Error}}{n} = \frac{0.026885 - 0.000752}{2} = 0.01306565$$

$$\hat{\sigma}^2_{\beta} = \frac{MS_{B} - MS_{Error}}{an} = \frac{0.160991 - 0.000752}{3(2)} = \mathbf{0.0267065}$$

R Codes to the Estimates of Variances:

```
> model_anova = anova(model)

> ms = as.vector(model_anova[[3]])
ms=ms[-1]

> n=2; a=10; b=3

> cat("Method of Moments Variance Component Estimate","\n",
   "Variance of Error = ", ms[3],"\n",
   "Variance of Interaction = ", (ms[2]-ms[3])/n, "\n",
   "Variance of Part = ", (ms[1]-ms[3])/(b*n), "\n")
```

Method of Moments Variance Component Estimate
Variance of Error = 0.0007516667
Variance of Interaction = 0.01306667
Variance of Part = 0.0267065

R Codes to Obtain REML Estimate:

```
> library(lme4)
> model2 = lmer(y~1+ Operator+ (1|Part)+(1|Part:Operator))
> summary(model2)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ 1 + Operator + (1 | Part) + (1 | Part:Operator)

REML criterion at convergence: -126.6

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.42306 -0.36626 -0.00037  0.37732  1.95386

Random effects:
 Groups         Name        Variance  Std.Dev.
 Part:Operator  (Intercept) 0.0130667 0.11431
 Part           (Intercept) 0.0223509 0.14950
 Residual                   0.0007517 0.02742
Number of obs: 60, groups:  Part:Operator, 30; Part, 10

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.79800    0.05983  13.338
Operator2   -0.02700    0.05185  -0.521
Operator3    0.02750    0.05185   0.530


Correlation of Fixed Effects:
            (Intr) Oprtr2
Operator2   -0.433
Operator3   -0.433  0.500
```

What Have We Learned?

We discussed fractional factorial design, specially one-half and one-quarter fractional factorial design in this chapter.

We gave examples for balanced and unbalanced design and discussed moment estimators, ML estimators and REML estimators.